# GREEN SYNTHESIS OF ZINC OXIDE NANOPARTICLES

**Cluster Innovation Centre**
**University of Delhi**

Dinesh Yadav | Divya Bandaru
Swarnim Sharma | Gaurav Jha

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Prof. Mahima Kaushik

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# CERTIFICATE OF ORIGINALITY

The work embodied in this report entitled "GREEN SYNTHESIS OF ZINC OXIDE NANOPARTICLES" has been carried out by Dinesh Yadav, Divya Bandaru, Swarnim Sharma, and Gaurav Jha for **"Engineering Chemistry**". We declare that the work and language included in this project report are free from any kind of plagiarism.

(Name and signature)

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# ABSTRACT

Zinc oxide nanoparticles (ZnONPs) have been studied for their potential in a variety of biological applications, such as biocompatibility, antimicrobial activity, and wound healing. In this study, ZnONPs were synthesized using a green method, which is more environmentally friendly than traditional chemical methods. The nanoparticles were characterized using analytical techniques, including UV–Visible spectroscopy and FTIR.

The interaction of ZnONPs with calf-thymus DNA (CT-DNA) was also investigated using UV–Visible Spectroscopy. The results showed that ZnONPs could interact with CT-DNA and stabilize it. This suggests that ZnONPs could have potential applications in biosensing and biomedical applications.

# INTRODUCTION

Nanomaterials are materials that have at least one dimension smaller than 100 nanometers (nm). This size reduction gives them unique properties that are not found in their bulk counterparts. For example, their band gap (the energy difference between the valence band and conduction band) can be widened, leading to a larger surface area and higher surface energy.

The field of nanoscience has attracted significant interest from researchers in a variety of disciplines, including biotechnology, physics, chemistry, environmental science, material science, and information technology. This is because nanomaterials have the potential to revolutionize many industries and improve our lives in many ways.

Metal oxides are a class of materials that are made up of metal ions and oxygen atoms. They are found in nature and are also synthesized in the laboratory. Metal oxides have a wide range of properties, including electrical conductivity, optical properties, and catalytic activity.

Nanotechnology has revolutionized the synthesis of metal oxides, leading to the development of new materials with improved properties. For example, ZnO nanoparticles (ZnONps) have attracted considerable attention due to their biological applications. ZnONps have been shown to have antibacterial, antifungal, and wound healing properties.

Various chemical and physical methods exist for synthesizing metal oxide nanoparticles, but the green synthesis approach is gaining popularity, particularly in the pharmaceutical and biomedical fields, due to the use of less harmful chemical reagents . Biosynthesis, involving plants or plant leaf extracts, is also gaining importance from an economic perspective, and recent advancements are focused on developing cost-effective synthesis methods . The use of plants or plant extracts in synthesis offers several advantages, including the production of metal oxide nanoparticles with diverse sizes and shapes.

Here are some of the reasons why green synthesis of zinc oxide nanoparticles is better than chemical methods:

- Environmentally friendly: Green synthesis methods use plant extracts, bacteria, or other biological materials as the reducing and capping agents. These materials are non-toxic and biodegradable, making them more environmentally friendly than chemical methods, which often use toxic solvents and reagents.
- Safer: Green synthesis methods produce nanoparticles that are less toxic than those produced by chemical methods. This is because the biological materials used in green synthesis methods do not contain any harmful chemicals.

- More controllable: Green synthesis methods are more controllable than chemical methods. This is because the reaction conditions can be more easily adjusted to produce nanoparticles with the desired size, shape, and properties.
- More efficient: Green synthesis methods are more efficient than chemical methods. This is because they use less energy and produce less waste.

The green synthesis of a diverse range of nanostructures has been extensively explored in recent decades as a more environmentally friendly and sustainable alternative to traditional methods. Plant-based nanoparticles have several advantages over traditional nanoparticles, including their biocompatibility, biodegradability, and tunable properties.

ZnO is a rare inorganic compound found in nature. It is typically found as a crystalline substance. Naturally occurring ZnO may have a classic red or orange hue due to manganese impurities. When purified, ZnO is a white crystalline powder that is nearly insoluble in water. ZnO is an n-type semiconductor with a wide band gap of 3.37 eV at ambient temperature, a high excitation binding energy (60 meV), superior chemical stability, a low dielectric constant, a strong electrochemical coupling coefficient, and excellent optical absorbance. ZnO NPs have been extensively used in a variety of applications, including textile materials, beauty products, diagnostics, solar cells, sensing applications, dielectric materials, electric actuators, piezoelectric elements, room-temperature UV lasing, and even microelectronics, due to their relatively low toxicity and size-dependent characteristics. For therapeutic purposes, ZnO has been shown to be more beneficial and efficient than other metals in the biosynthesis of NPs. Several studies have demonstrated that plant extracts can be used to synthesize ZnO NPs. The size and properties of the nanoparticles can be controlled by varying the plant extract, the reaction conditions, and the method of synthesis.

The green synthesis of ZnO NPs is a promising new method for the production of these versatile and functional materials. This method has the potential to reduce the environmental impact of nanoparticle production and to open up new applications for ZnO in the fields of biology, medicine, and technology.

# MATERIALS AND METHOD

(i)Materials Required

Different plant leaves for phytochemicals, zinc acetate, NaOH, cotton cloth, sterilized tip, lab funnel, magnetic stirrer, heat oven, aluminum foil, glass rod, spatula, Tripod stand, Eppendorf, gloves, ethanol, Whatman filter paper, beakers, and conical flask.

(ii) Methods

**UV–Visible Spectrophotometric studies**

UV–Visible spectra were taken from Shimadzu UV–Visible spectrophotometer (UV-1650 PC). Desired concentrations of CT-DNA were obtained by mixing appropriate amounts of DNA in Tris-EDTA buffer (pH 7) and were preserved below 4 °C. Interaction studies between CTDNA and ZnONPs were done by the measurement and analysis of UV–Visible spectra. The spectroscopic changes in absorbance were recorded by keeping constant DNA concentration (100 μM) and varying the concentration of ZnONPs, in the wavelength range of 220–320 nm.

**FTIR studies**

FTIR spectra were recorded by using Thomas Scientific Spectrometer. The spectral analysis provides information of the presence of functional groups in the samples.

# PROCEDURE

1. The collected neem leaves samples were thoroughly washed several times with doubledistilled water.

2. Dried leaves were mixed with 100 mL of double-distilled water and heated at 60°C for 1 hour using a magnetic stirrer until the solution turned deep white in color.

3. The solution was allowed to cool to room temperature. The extract was filtered using Whatman no. 2 filter paper and stored at 4°C for further use.

4. 43.9 g of zinc acetate was dissolved in 50 mL of distilled water with constant stirring. To adjust the pH of the solution to 12, NaOH (0.4 g dissolved in 50 mL) was used.

5. After the mixture became clear, 5 mL of *A. indica* was added. The solution was slowly heated at 60°C overnight.

6. The paste was calcinated for 2 hours at 400°C.

7. A pinch of powder was mixed with 15 mL of water and sonification was performed for 15 minutes.

8. The sample was taken in the required quantity and characterization techniques were performed.

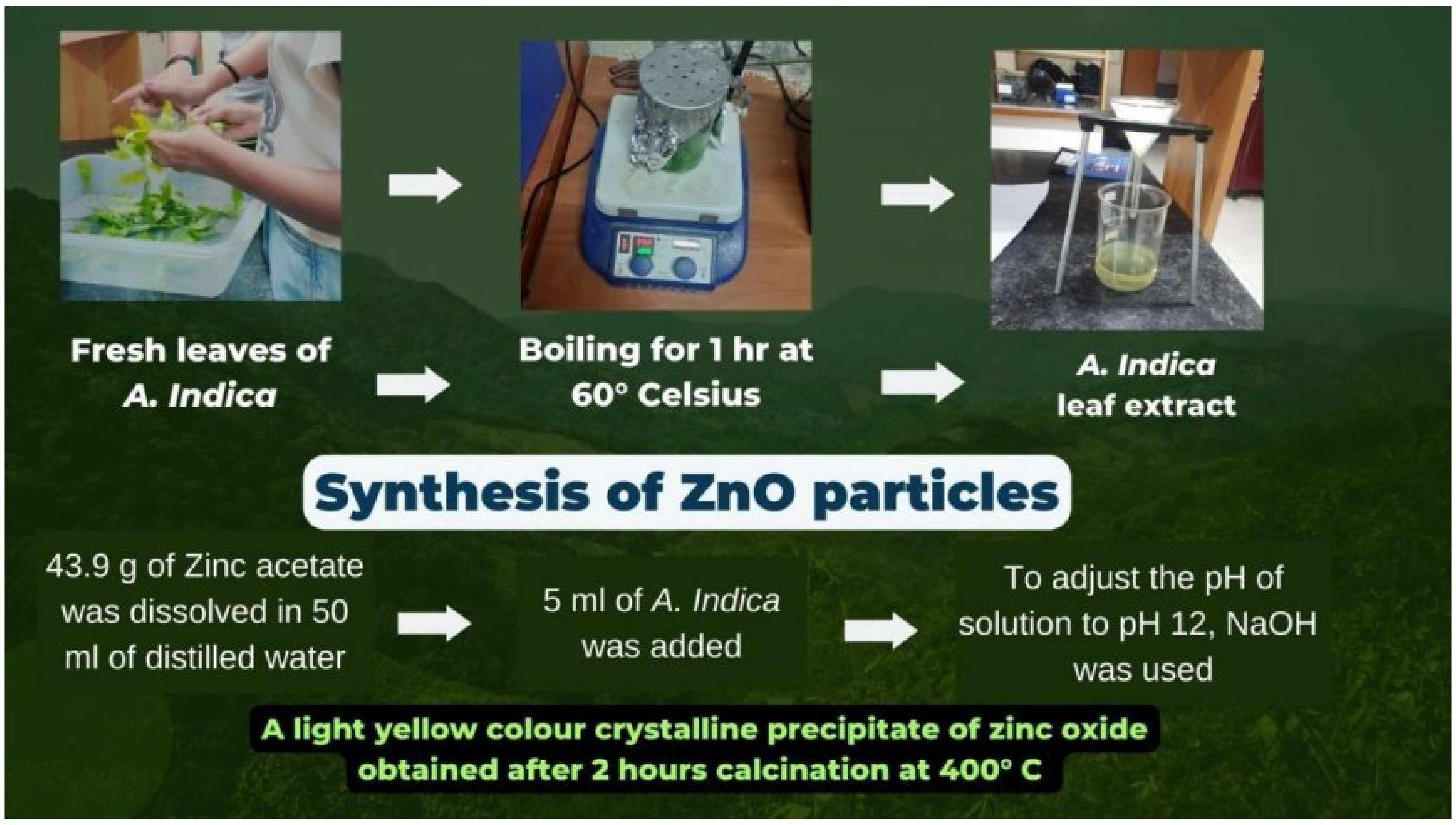


Fig 1a. Schematic representation of the preparation of *Azadirachta Indica* leaf extract and synthesis of ZnO nanoparticles

# CHARACTERIZATION TECHNIQUE

## (i) UV–Visible spectroscopic studies

UV-Vis spectroscopy is an ideal method that is usually performed for the confirmation and characterization of the synthesis of ZnONPs based on surface plasmon resonance (SPR). The UV–Visible absorption spectra of ZnONps synthesized in the presence extract were recorded at room temperature and are shown in Fig. 2a. The optical absorption spectrum of the synthesized ZnO NPs was determined in the range of 200–800nm. The UV–Visible spectra were recorded for the dispersed solution prepared by ZnONPs in water. ZnONPs with extract show sharp peaks in UV–Visible spectrum around 379.5 nm (neem) 361 nm (Tamarind) and 377 nm (Tulsi) confirming their synthesis.

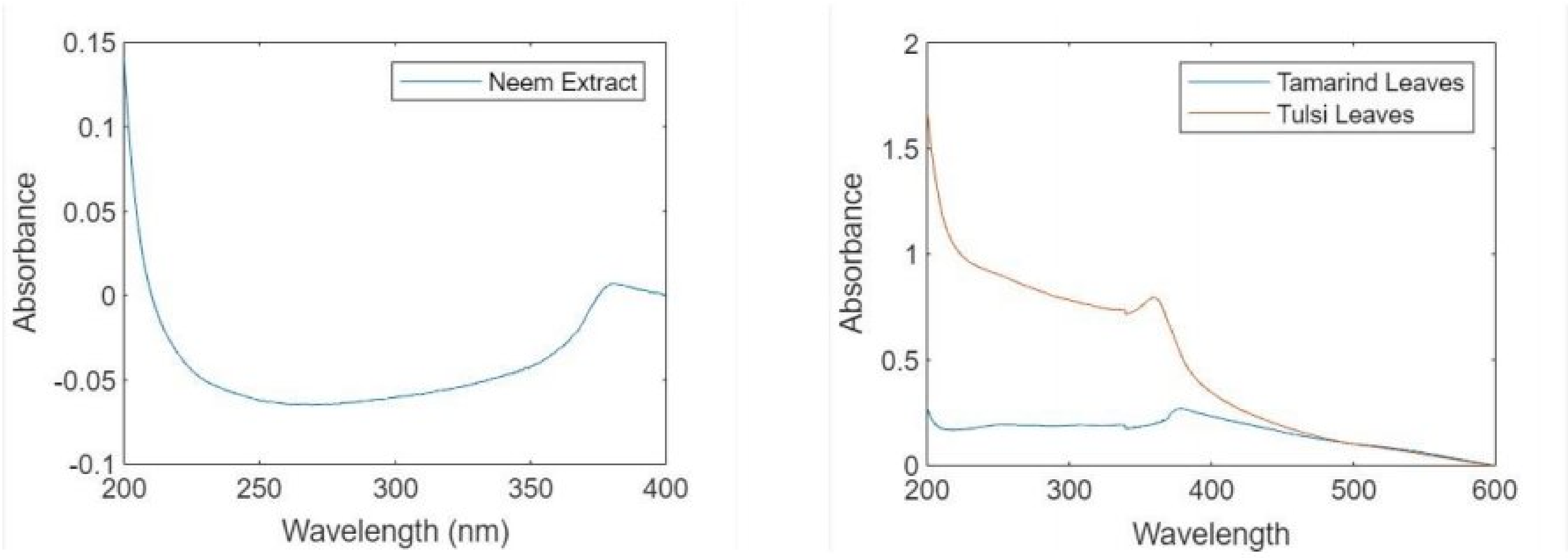


Fig 2a. UV-Visible absorption Spectra of ZnONPs (1.5mM) in presence of increasing concentrations of CT-DNA (100-200μM)

UV–Visible Spectroscopy is an effective method for understanding the type of binding of DNA with nanoparticles. Spectroscopic titration of CT-DNA with nanoparticles was performed to investigate the possibility of binding of ZnONPs to DNA. The spectra were recorded for samples in which the DNA concentration was kept constant and the nanoparticle concentration was varied. It has been reported that a hypochromic shift in the absorption peak of DNA at 260 nm, upon interaction with nanoparticles, indicates binding through intercalation or electrostatic interaction, whereas a hyperchromic shift indicates the breaking of the secondary structure of DNA.

In this case, Figure 3a shows the changes in the UV spectra of ZnONPs (constant concentration) with increasing concentrations of CT-DNA (100–200 μM). The absorption spectra showed a red shift, which indicates the binding of nanoparticles with DNA, resulting in an increase in the size of the nanoparticles. Likewise, Figure 3b shows the UV-spectra of CT-DNA alone and at different concentrations of ZnONPs (0.5–2.0 mM). The decrease in the intensity, or hypochromic shift, after the addition of a constant concentration of nanoparticles without any shift indicates the direct formation of a complex.

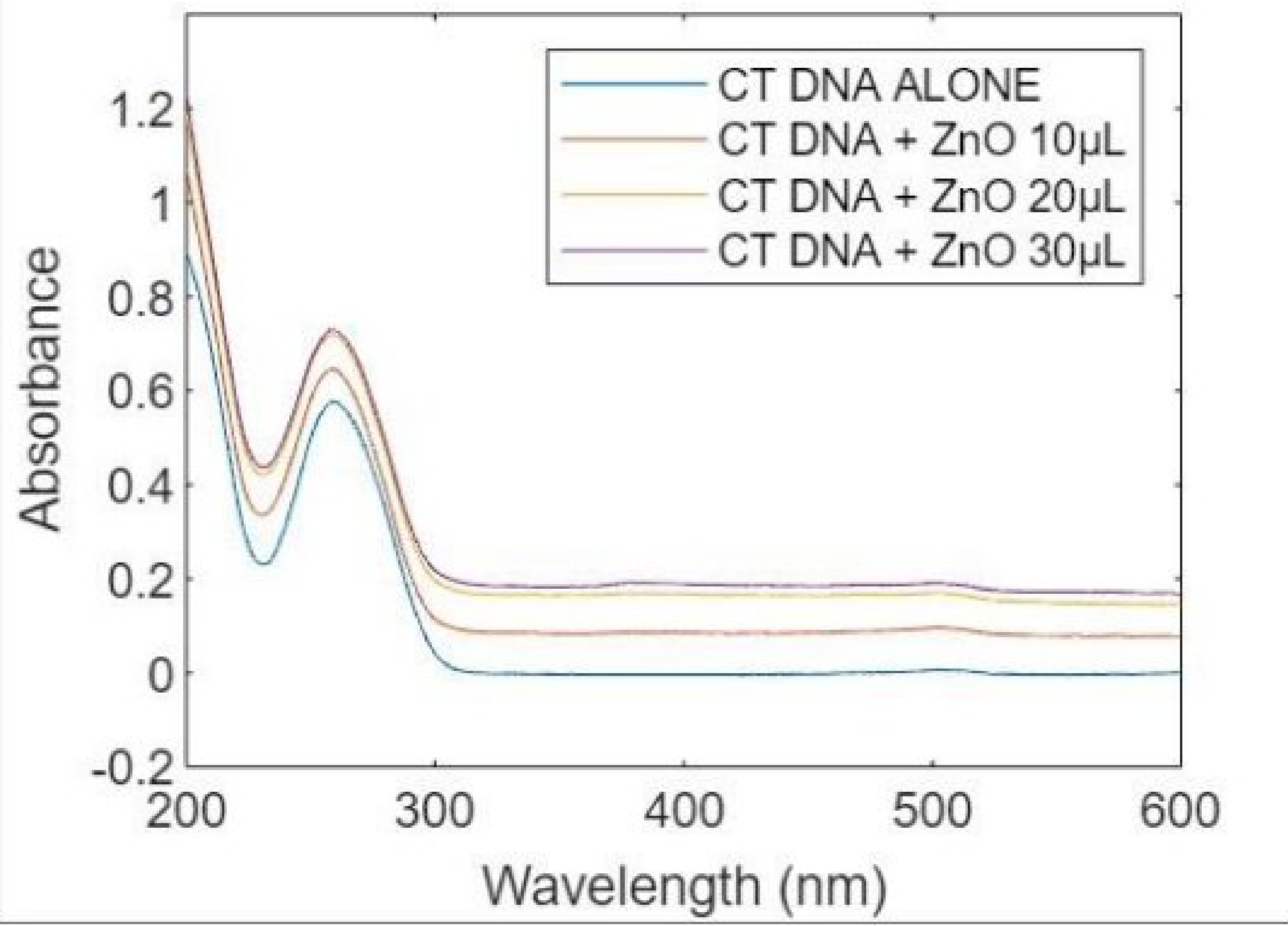


Fig 3a: UV-Visible absorption spectra of CT-DNA (100μM) in presence of Increasing concentrations of ZnONps (0.5-2.0mM).

(ii) FTIR

FTIR spectrum analysis was performed to determine the surface chemistry of the ZnONPs and applied to obtain the details of functional groups involved in the biomolecules responsible for reducing and capping the bio-reduced ZnONPs. The FTIR spectrum results revealed a complex of bioorganic compounds in the plant extract, which are responsible for the reduction and stabilization of ZnO NPs. The peaks indicate the characteristics functional group present in the synthesized zinc oxide nanoparticles. It is inferred that the samples have absorption peaks in the range of

SAMPLE 2

- The absorption peak at 418.84 to 576.716 cm-1 corresponds to metal-oxygen (ZnO stretching vibrations) vibration mode.
- The peak at 1045.42 cm-1 is ascribed to the stretching vibration of C-N bond of the primary amine or to the stretching vibration of the C-O bond of the primary alcohol.
- The peak at 1246.71 cm-1was ascribed to primary alcohol in-plane bend or vibration.
- The additional peaks observed at 1448 cm-1 may be due to the O-H bending and stretching frequency of H2O molecules.

SAMPLE 4

- The absorption peak at 403.4 to 588.34 cm-1 corresponds to metal-oxygen (ZnO stretching vibrations) vibration mode.
- The peak at 1045.42 cm-1 is ascribed to the stretching vibration of the C-N bond of the primary amine or to the stretching vibration of the C-O bond of the primary alcohol.
- The peak at 1078.71 cm-1 was ascribed to primary alcohol in-plane bend or vibration.

- The additional peaks observed at 1560 cm-may be due to O-H bending and stretching frequency of H 2O molecules.

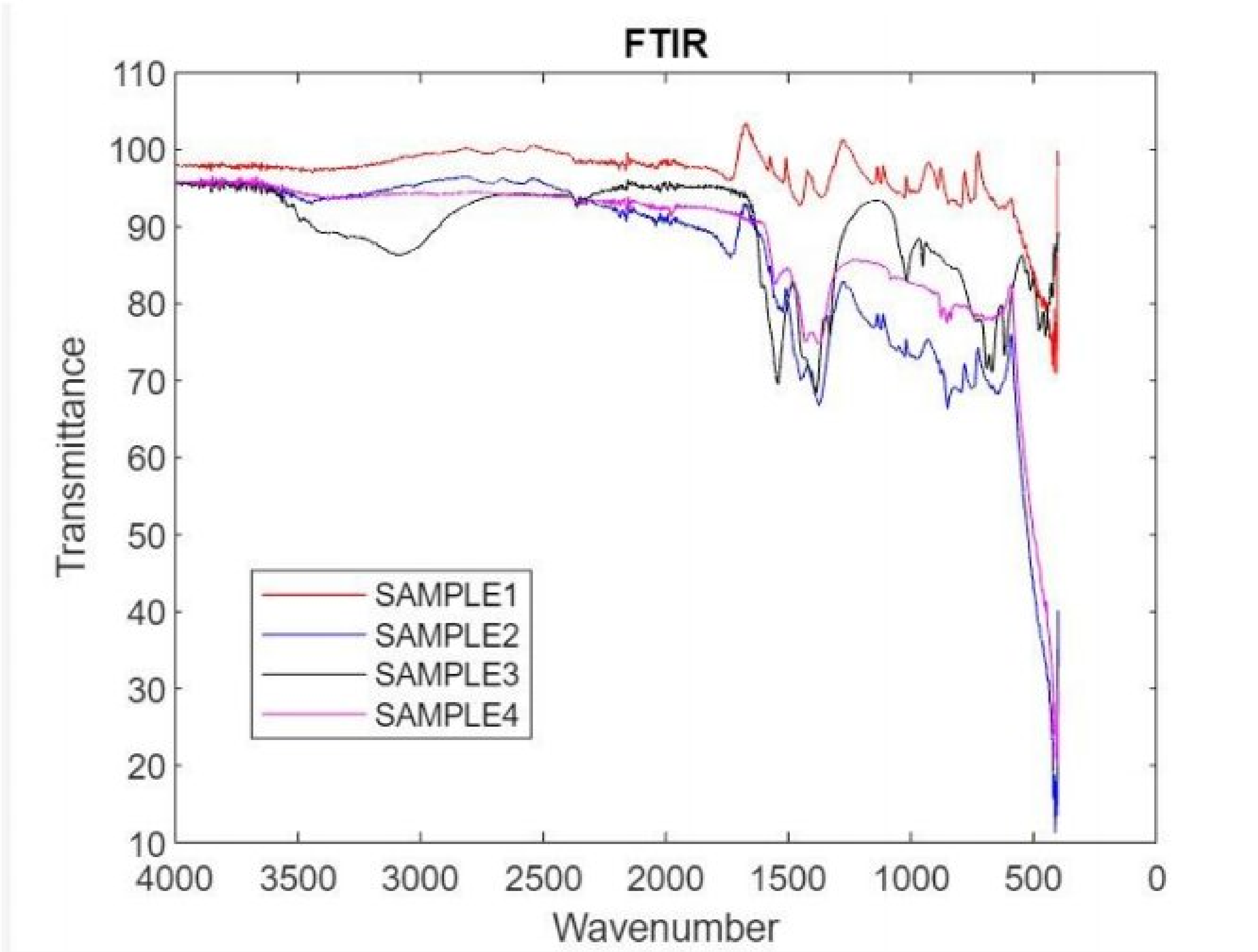


Fig 4a: FTIR spectra of green synthesized ZnONps

# Conclusion and Discussion

ZnO nanoparticles were successfully synthesized by the green synthesis method using Zinc Acetate and neem leaves. This report focuses on the green synthesis, characterization and interaction studies of ZnONps and CT-DNA using physicochemical techniques.

In FTIR spectroscopy, pure ZnO nanoparticles showed stretching vibrations and peaks were matched with standard fictional groups of ZnO. Thus, these interaction studies suggest that thus prepared ZnONps can be used for the formation of

DNA-Nanoconjugate complex for using them in bio-medical applications.

Apart from the above-mentioned studies, some *in vitro* and *in vivo* studies are required to demonstrate the biological and biosensing applications of the metal oxide nanoparticles.

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